Physiological response prediction based on Uncertainty-Awared Attention

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Abstract

Physiological response prediction is important for health monitoring. Some works has been done in this area to make predictions more accurate, while there are still many works to be done to improve the reliability of the prediction. In our work, We build UAP, a regression model implemented with LSTM and uncertainty-awared attention to not only make prediction but also output prediction reliability. Then, we utilize this model to analyse Fabian dataset to see prediction accuracy and reliability.

1 Introduction

Through wearable health monitoring devices, we can collect a lot of data about human body physiological response and the environment. For example *AirBeam* can collect environment data, like temperature, humidity etc. *BioHarness* can collect breath rate and heart rate, *Empatica E4* can capture human movement data.[1] These data are relevant with each other making it possible to predict unknown data with data already known. Models can be built to model connection between known data to predict unknown data.

Some works have been done in the work of physiological prediction. *Farrokh* et al. [2] builds a frame of human physiological response prediction using Support Vector Machine(SVM). *Farrokh* et al. [3] also builds SVM model to predict physiological response of milkcow. Also, some physiological response prediction models are implemented with LSTM, which helps improve model's ability in modeling semantic information through long sequences.

But LSTM is not good enough for feature modeling and interpreting, which can be solved by adding attention mechanism. Attention mechanism is very useful in not only modeling relevant features but also interpreting features. It has achieved great success in areas like image recognition and machine translation. While, there are still some drawbacks of attention mechanism. Basically, attention mechanism is a method to assign weight to sequence elements. That makes the supervised learning more weakly-supervised, which decrease the reliability of the model. This is sever in safety-critical areas, like health monitoring.

Fortunately, some methods have been put forward to output prediction reliability. *Alex* et al. [5]

talks about uncertainty needed in the area of computer vision. *Jay* et al. [4] presents a method to allow the model to output uncertainty on each features.

In this work, we present UAP(Uncertainty-awared Attention Predictor), which not only utilize attention mechanism and LSTM to better interpret input sequences but also output the uncertainty of the prediction. We achieved our model by turn the UA model presented by *Jay et al.*[4] into a regression model. Then we utilize the UAP model to make prediction based on the Fabian dataset. We also present two baseline(LSTM and LSTM+Bayesian) to compare with the UAP. After that, we analyse the output to see the prediction accuracy and uncertainty.

It turns out that the RMSE of UAP model is 3.7, RMSE of baseline model is 2.7, so the performance is not better than the baseline. We think the problem may be as follows:
1.The UAP model is not tuned well enough 2.Some more modification should be done to change the original classification model into a regression model.

2 Related work

LSTM(Long Short Term Memory):
LSTM is a special kind of recurrent neural network, which is can overcome the drawbacks of standard neural network and can deal with the limitation of standard neural network. There are mainly two limitation of standard neural network. Firstly, when the input and output are of different length, problems, standard neural network can not work. Secondly, the dependency of data can not be modeled. With an hidden state variable, recurrent neural network can pass the variables of former element to latter ones to model the dependency of sequence data. While ordinary RNN preforms badly in modeling long sequence data, so the LSTM(Long Short Term Memory) is introduced by adding additional c variable to control whether to pass variables. With this additional variable, the network can avoid the former information from vanishing in the long transferring process, so can model long sequences well.

Attention Mechanism:
Attention Mechanism has vast area of application, I have been applied to the area of machine translation, image recognition, machine reading comprehension. It shows great ability and potential in modeling sequence data. Some works have been done to use attention mechanism for interpretability, as Choi et al. [6] which combine RNN and attention to provide attention on both hospital visits and variables.

Prediction reliability:
Although delicate neural networks show great effect in many areas and works, model reliability is still a big problem. Some works has been done to output prediction reliability. When prediction reliability is not high enough, this prediction will be abandoned, so models can make more accurate and reliable predictions. One way to achieve this goal is using Bayesian models to calculate predictive uncertainty. In some works, the uncertainty is used as a parameter for dropout calculation.

3 Experiment

3.1 Dataset

We utilize the Fabian dataset in our experiment.

Fabian dataset is collected from a healthy, 30-35 year old male subject, over a period of four days. The subject was asked to wear some sensors (E4, SensorTag, BioHarness, Airbeam) the E4 on the left wrist, the SensorTag was on the right ankle, the BioHarness was located on the accompanying sports shirt, the Airbeam was located on the back. After that, the subject performed different exercises. 7.0-9.0 hours of data, with average of 8.0 hours is recorded everyday.

The AirBeamis a portable environmental sensor. This sensor can measure humidity, temperature, environmental sound level. The BioHarnessis capable of measuring physiological response, like heart rate, breath rate etc. The E4 is a wearable wristband capable of measuring blood volume pulse, heart rate, humidity excitement level etc.The SensorTag is a combination of sensors, including magnetometer sensor, accelerometer, humidity sensor, ambient temperature sensors etc.

3.2 Data processing

we used accelerometer features from the wrist and arm and breath rate for our model.We fifirst concatenated all the IMU features and breath rateacross our training days to one array. Then we performed a normalization on that array. Next, we performed a sliding window for each day with 10-time steps in a window. Afterward,we do the normalization with the mean and variance we got previously and apply to those sliding window.

Then we split the input data into three sets: train data, validation data, evaluation data. These data are all processed into matric.The shape of train data is(14000, 10, 43), the validation data is (5333, 10, 43), the evaluation data is (5098, 10, 43). The first dimension is the number of records, the second one is the length of every record, which corresponds to the sliding window size, the third dimension is the number of record features.

3.3 Models

1) Baseline1: LSTM

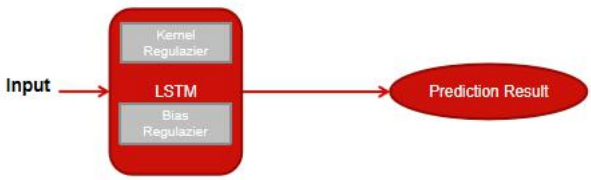

Figure 1

The first baseline is built with LSTM. Figure 1 shows the design of baseline1.

We had our LSTM layer with recurrent dropout = 0.01, dropout = 0.01; kernel regularizer = l2(1e-15) and bias regularizer = l2(1e10). The reason why we had regularizers is that the regularizewould ensure our weight and bias are not too large and prevent overfitting.

2) Baseline2:LSTM+BNN

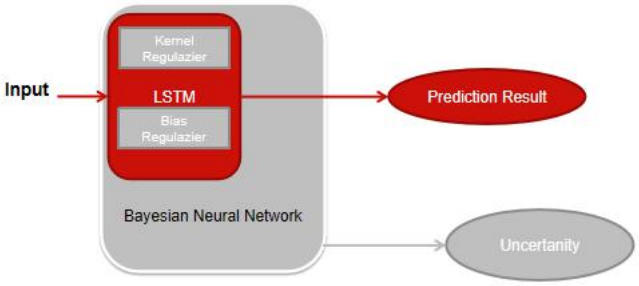

Figure 2

The second baseline is built based on the first baseline. Figure 2 shows design of the model.

We model epistemic uncertainty using Monte Carlo dropout. The LSTM architecture places dropout with p =0.

3) UAP(Uncertainty-Awared Attention Predictor)

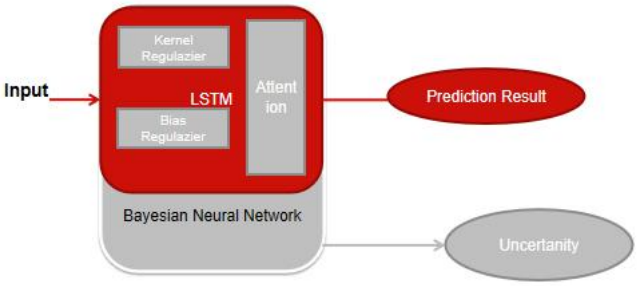

Figure 3

Figure 3 shows design of the UAP model.

We build our model by changing binary classification model, RETAIN presented in the [4] to an regression model. To complete the change we do some modification to the original model:

1. In the model evaluation part, we change the accuracy to RMSE(Root Mean Squared Error)
2. In the model output part, we change the activation function of the last layer from sigmoid to linear.
3. In the loss function part, we change the loss function from sigmoid cross entropy to MSE(Mean Squared Error)

4 Result

We present our result as follows:

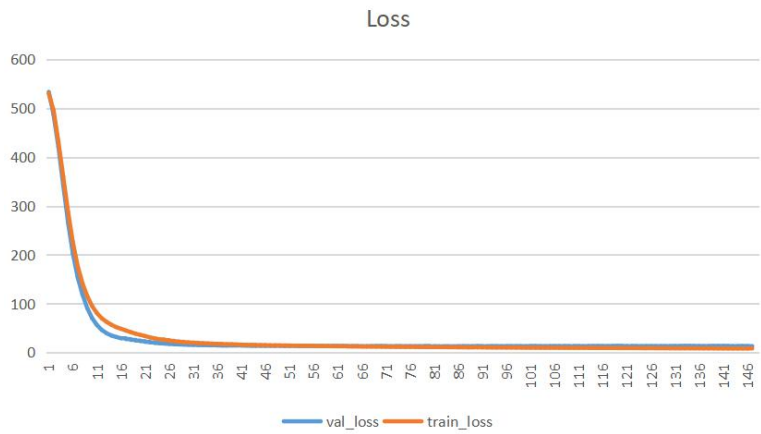

Figure 4

Figure 4 shows loss function of both training and validation. From the figure, we can see the loss function has been stable after training.

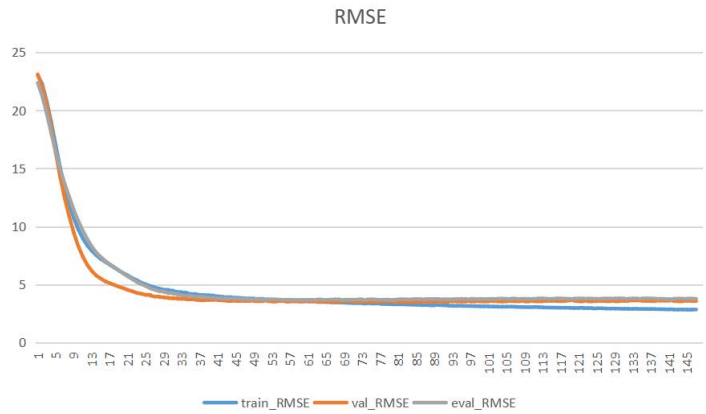

Figure 5

Figure 5 shows RMSE of the model. After 90 epochs, the model seems to be a little overfitting. But the RMSE of validation and evaluation dataset have become stable. And the value is around 4.

Epoch	1	20	40	60	80	100	120	140
Loss	500.759	36.273	15.411	14.022	14.001	14.255	14.539	14.185
RMSE	22.376	5.938	3.835	3.684	3.706	3.752	3.797	3.754

Table 1

Table 1 shows the loss and RMSE of evaluation after different training epochs. Before epoch 100, the loss and RMSE keep dropping, after that the loss and RMSE rises. The model has been overfitting after that. So we choose the RMSE of epoch 80 as our final result.

Then we compare the result of baseline and UAP.

	LSTM	LSTM+BNN	UAP
RMSE	2.89471415	2.485112479	3.768299737

Table 2

Table 2 shows the final results of two baselines and UAP.

Compare baseline1 and baseline2, we find uncertainty-awared LSTM outperforms single pure LSTM model. Making use of model uncertainty is helpful in improve model performance.

Compare baselines with UAP, we find RMSE of UAP is lower than both baselines.The explanations for that are in two ways:

1) It is not very accurate to measure the attention uncertainty. So reliable prediction may be regarded as unreliable, so it will be abandoned. Some unreliable prediction may be regarded as prediction, so it will remain and decrease the RMSE eventually.

2) The model hasn't be tuned good enough to outperform the baseline. Hyper parameters, like lambda in loss function, dropout prob.

5 Conclusion

In our work, we build three models and utilize Fabian dataset to test our models. We prove it is possible to combine attention and uncertainty in regression physiological response prediction. The UAP model is able to predict Breath rate with accelerator data and former breath rate, and get the result of 3.7RMSE, which is lower than the baseline. We proposed two ways of explanations, and in the future, more works can be done to explore these explanations.

More works can be More exploration works can be done to improve the performance of the model.

Some works can be done to improve the performance of the model:

1.Some further fine-tuning of the hyper parameters, like lambda in the loss function.

2.Introduce other kind of attention, which is more suitable for regression physiological response prediction problems.

6 Future work

Some works can be done to explore the explanations.

For explanation 1, we can remove the attention mechanism in the UAP model and compare performance of attention-removed UAP with UAP to see whether the problem lay in the uncertainty part.

For explanation 2, Some works can be done to improve the performance of the model:

1.Some further fine-tuning of the hyper parameters, like lambda in the loss function.

2.Introduce other kind of attention, which is more suitable for regression physiological response prediction problems.

Besides these, more exploration works can be done:

In our work, we use the data of accelerator and former breath rate to predict latter breath rate. Wile there are still some data haven't been used, like heart rate, environment temperature, humidity etc. It will be helpful if we can make use of these data to make predictions.

Also, although this model is designed for physiological response prediction, it can be applied to many other time series, like stock price. Some works can be done to apply this model to other time series data.

7 References

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